

LINER VARIABLE DIFFERENTIAL TRANSFORMER (LVDT) (3)

This is the most widely used inductive transducer for translating linear motion into an electrical signal. The basic construction of an LVDT is shown in Figure below.

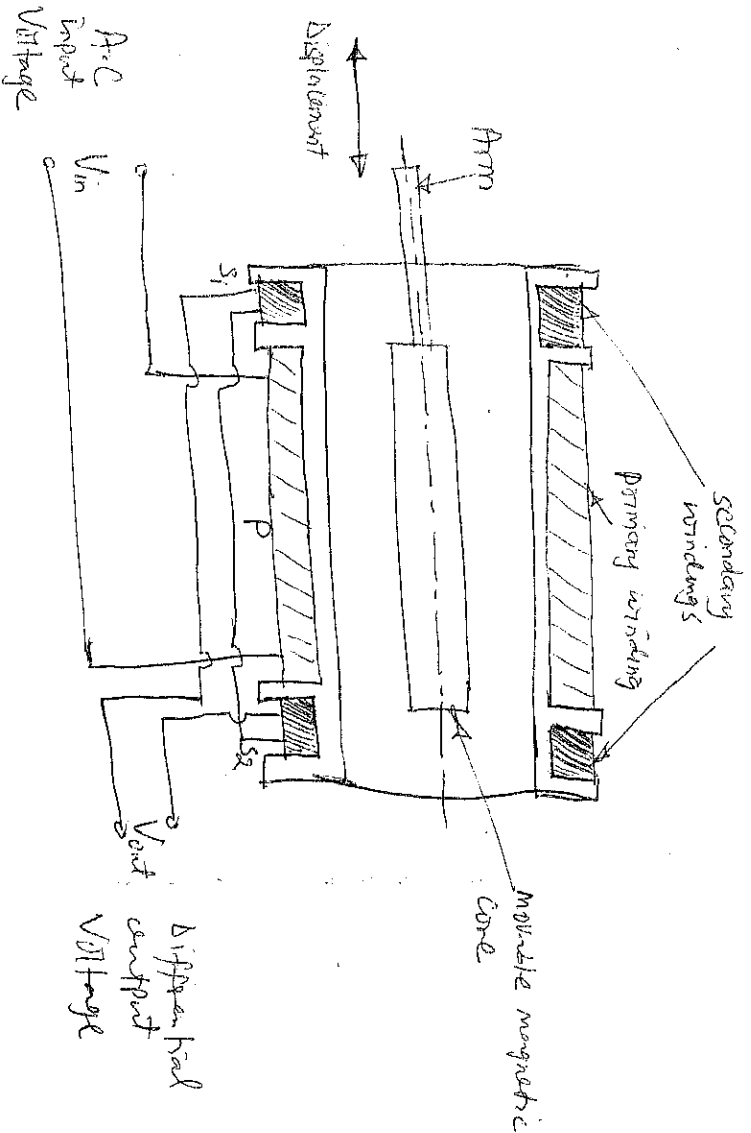


Figure LVDT

LVDT is a differential transformer consisting of one primary winding P and two identical secondary windings S_1 and S_2 , wound over a hollow bobbin of non-magnetic and insulating material. The secondary windings S_1 and S_2 have equal ^{Number} windings of turns and are arranged concentrically and placed either side of the primary winding P . It consists of a soft iron core, which is attached to the sensing element of which displacement is to be measured, is the slope of rod or cylinder slides freely in the hollow portion of the bobbin. ~~The~~ primary windings is connected to an a.c. source voltage varying from 5 to 25V and of frequency ranging from 50Hz to 20kHz. When the core is moved into the bobbin it varies coupling of primary winding to secondary windings S_1 and S_2 . In null position of the core i.e. in central position, coupling of primary winding to both of the secondary windings are equal and so output voltages

induced in secondary windings S_1 and S_2 are equal. As the core is moved towards left from its null position, the magnetic linkages to secondary windings S_1 increases and to secondary windings S_2 decreases because the portion of the core inside secondary winding S_1 increases and that inside secondary S_2 decreases. Therefore, output voltage induced in secondary winding S_1 increases whereas the output voltage of the core to the right have opposite effect. Secondary windings S_1 and S_2 are connected in series opposition so that difference of output voltages of secondary windings gives the measurement of displacement.

ELECTRONIC RECORDING DEVICES

In most of the industrial and research processes continuous monitoring of the condition, state, or value of the process, variables (such as force, temperature, flow, humidity, voltage, current, etc) is necessary. All these needs can be done by electronic recording devices, called the recorders.

RECORDING - A recorder is a measuring instruments that displays a time varying signal in a form of easy to examine and re-examine even after the original signal has ceased to exist.

Recorders generally provide a graphic record of variations in the quantity being measured, as well as an easily visible scale on which the indication is displayed. A recorder records electrical and non-electrical quantities and a function of time or related time signals to each other. Electrical quantities such as current, voltage etc. can be recorded directly while non-electrical quantities such as pressure, temperature, speed, flow etc. are recorded indirectly by first converting them into the form of electrical signal with the help of sensor or transducers.

TYPES OF RECORDING DEVICES

Recording devices are of two types as follows:

- (i) Analog recorders
- (ii) Digital recorders

Analog recorders are of graphic recorders, oscillographic recorders and magnetic tape recorders.

Digital recorders are pressure recorder, flow recorder, speed recorder, liquid level recorder etc.

GRAPHIC RECORDERS

A graphic recorder may be defined as an instrument which ^{draws} a graph that relates two or more variables to time or to each other. A graph is an analog representation of discrete information.

Types of Graphic Recorders

Three types of graphic recorders are:

- (i) Strip chart recorders
- (ii) Circular chart recorders
- (iii) X-Y recorders

1. STRIP CHART RECORDER

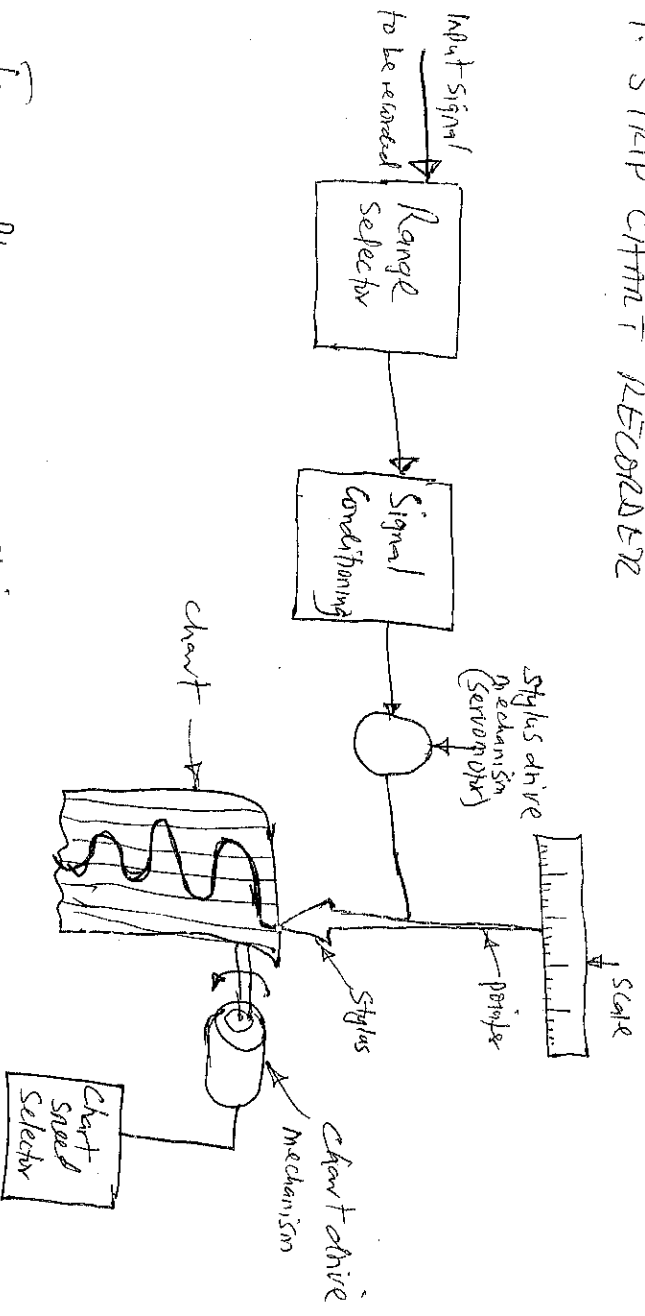


Figure Block diagram of a Strip chart Recorder.

The functional elements of a strip chart recorder are a Stylus (Recording Pen) for tracing the signal on the graph paper, a system which drives the ^{stylus} paper, a long roll of graph paper, a mechanism which drives the paper in vertical direction. The speed of a movement of graph paper (chart) is selected through a speed selector which is connected to the chart drive mechanism. The strip chart recorder also consists of a range ~~selector~~ switch and a signal conditioning circuit at its input section.

So that the level of the input signal is within the limit acceptable by stylus drive system.

When the input signal to be recorded is fed to the strip chart recorder the input signal is processed through a range selector and signal conditioning and then it is applied to the stylus drive system. The input to the stylus drive mechanism is a voltage or current signal corresponding to the signal to be recorded. The stylus drive mechanism is driven by a servomotor. The stylus drive mechanism moves the stylus horizontally by an amount proportional to the magnitude of the input signal. Simultaneously the chart driving mechanism moves the graph paper vertically at a certain speed selected through the chart speed selector. The graph paper is usually moved at speeds of 1-100 mm/s.

The simultaneous movement of the stylus and chart in horizontal and vertical directions respectively traces (records) the input signal with respect to time on the graph paper. The strip chart recorder also uses a printer connected to the stylus. The printer moves over a uniformly marked scale thus displaying the instantaneous values of recorded quantity.

2. CIRCULAR CHART RECORDER

In this recorder, a circular chart is used which has concentric circles ruled on it to form its scale. In addition, there are printed arcs extending from the centre of the chart to the paper edge. As the pen of the recorder is moved, it swings along these arcs. These arcs are called ^{time} arcs. The speed of the rotation of chart is usually one revolution per 24 hrs or for 7 days or any other speed which can be conveniently obtained by using synchronous motor with suitable gear assembly. The radial position of the pen at any time indicates the instantaneous value of the given quantity and measurement. The recording assembly is as shown in figure below.

Circular chart recorders are particularly suitable for direct action by a number of mechanical sensors such as bellows, load cells etc.

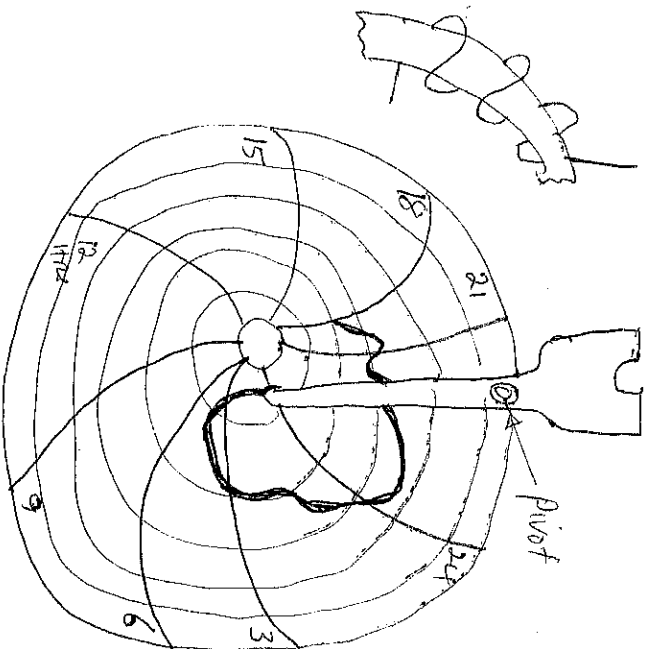


Figure Circular chart recorder

Magnitude of several variables can be recorded on a single chart which can be easily and conveniently analyze.

Chart diameter is limited to a maximum of 0.3m, speed of the chart is also limited and the chart do not run for a long period.

X-Y RECORDERS

These instruments are employed where it is desired to plot the relationship between two variables, $y = f(x)$ instead of plotting each variable separately as a function of time. These recorders display two separately varying quantities on the X and Y axes of Cartesian coordinates.

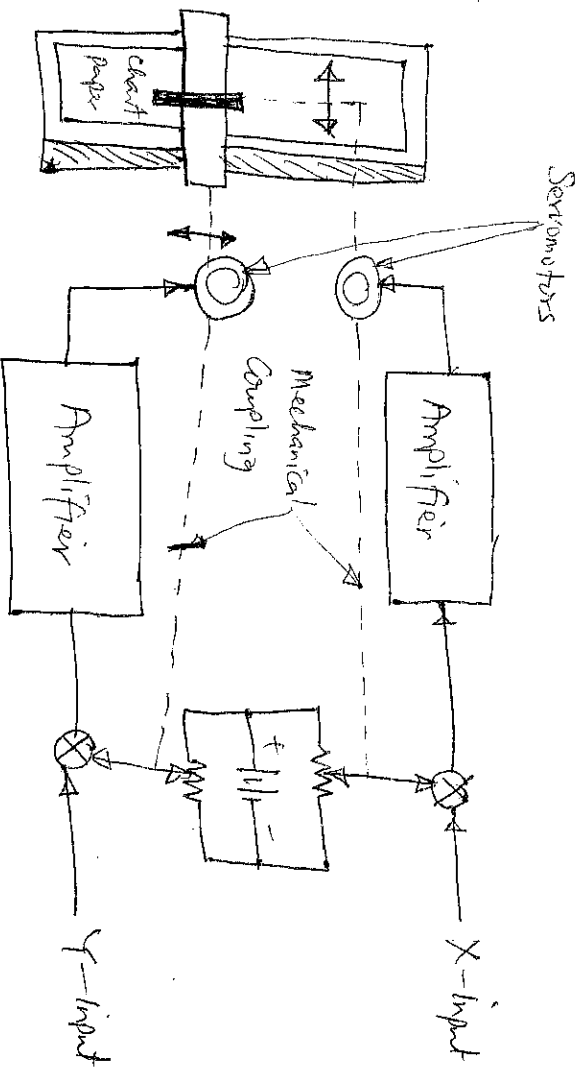


Figure X-Y Recorder

In such a recorder one of the variables is applied to the X-input and the other to the Y-input, and the recorder plots their variations against one another. In such recorders the chart remains fixed in its position and the pen moves simultaneously in both the X and Y directions in response to electrical signals applied to its two input terminals. Most X-Y recorders have a time base, so they can be employed to plot the variation of one variable against time as well.

Normally, X-Y recorders use pen and ink ~~marking~~ ^{recording} system and low cost paper for the chart. The mechanism for the movement of the pen in both X and Y directions is loop servo-motor ~~system~~ ^{system} to produce the self-balancing.

The rotation of the servomotor moves the mounting pan with the help of a string and pulley arrangement.

X-Y records are used for plotting current vs voltage curves of diodes and transistors, plotting B-H curves of magnetic materials and plotting speed-time curves for electric motors.